Predicting Risk Among Shelter Animals

Data Analysis: Louisville Metro KY – Animal Service Intake and Outcome

The Problem Context

Intake

Context: Shelters face chronic overcrowding and limited operational resources.

Critical Point: The exact moment of Intake often determines an animal's entire future trajectory.

The Problem: High-risk animals are frequently identified too late for effective, life-saving intervention. Euthanasia, death, or disposal are strictly classified as **At-Risk**.

Existing Analytical Methods in Shelters

Shelter Analytics

Predictive Classification (SVM, AdaBoost)

Predicting adoption chances and identifying euthanasia risk factors at intake.

Length of Stay Regression

Predicting exact days an animal will stay to optimize cage space and resources.

Time Series Forecasting (ARIMA)

Predicting intake volume for staff and inventory management during peak seasonal periods.

Deep Learning (ANN, LSTM)

Identifying complex behavioral and health patterns that traditional linear models might miss.

Natural Language Processing (NLP)

Analyzing pet descriptions on adoption sites to accelerate adoption rates.

The Louisville Dataset: Defining the Target

The Scope

The dataset tracks comprehensive animal service intake and outcome records from the Louisville Metro Animal Services (LMAS). This systemic, micro-level data serves as the raw material for predictive modeling, capturing the critical attributes of every animal crossing the threshold.

The Target Variable Definition

At-Risk

Euthanasia, Died, Disposal, Hospice

Synthesizing the Data: Feature Engineering Signals

Feature Parsing

Raw categorical text data—such as breed variations, intake conditions, and color descriptions—are rigorously cleaned, standardized, and grouped into distinct analytical categories to reduce noise and isolate structural patterns.

Recidivism Tracking

Tracking 'First Time' versus returning animals. Extracting engineered features that flag if an animal has a previous history with the shelter system, which serves as a known compounding risk factor for adverse outcomes.

Temporal Signals

Isolating time-based risk variables. Engineered features like 'Season' or intake month capture how shelter capacity strains during high-intake seasonal periods directly affect outcome risks and structural triage limitations.

Biting History

Extracting strict behavioral flags. Parsing intake notes and historical logs to isolate a definitive 'Bites' binary feature, which heavily correlates with complex medical needs or advanced behavioral triage requirements.

Project Architecture & Goals

Primary Objective

Early, binary classification of **At-Risk** versus **Safe** cases at the exact moment an animal crosses the intake threshold.

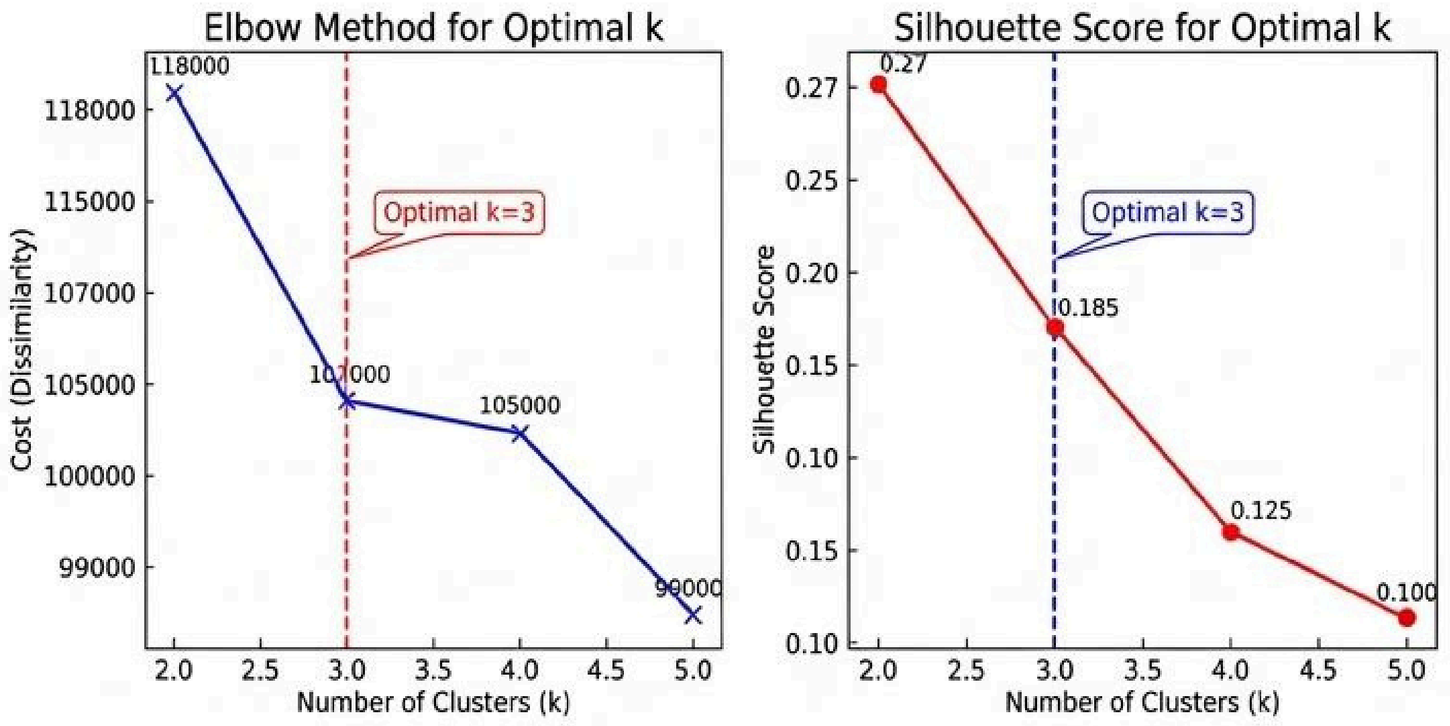
Uncovering Profiles

Discovering hidden risk groups within the broader shelter population using unsupervised algorithmic clustering.

Architectures Comparison

Evaluating and optimizing multiple models (Logistic, Tree-based) with a strict priority on the PR AUC metric.

Unsupervised Risk Profiles: K-Modes Clustering



K-Modes Clustering: Standard K-Means fails on highly categorical shelter data. K-Modes provides robust similarity matching specifically designed for these variable types.

Mathematical Consensus: Both Cost Dissimilarity and Silhouette Scores identify an inflection point, leading to the selection of optimal k=3 clusters.

Outcome: Achieved clear population stratification, isolating distinct structural clusters with significantly high and low mortality rates.

The Testing Sandbox: Experiments & Evaluation

The Environment

Data Split:
80% Training / 20% Testing holdout to prevent data leakage.

Imbalance Handling:
Utilization of SMOTE (Synthetic Minority Over-sampling Technique) to structurally balance the severe underrepresentation of At-Risk cases in the training data without losing critical signal.

Models Tested:
LightGBM, CatBoost, Random Forest, MLP, Logistic Regression.

The Measurement Matrix

PR AUC

Focusing on the trade-off between precision and recall, critical for highly skewed datasets to prioritize correct identification of the minority 'At-Risk' class.

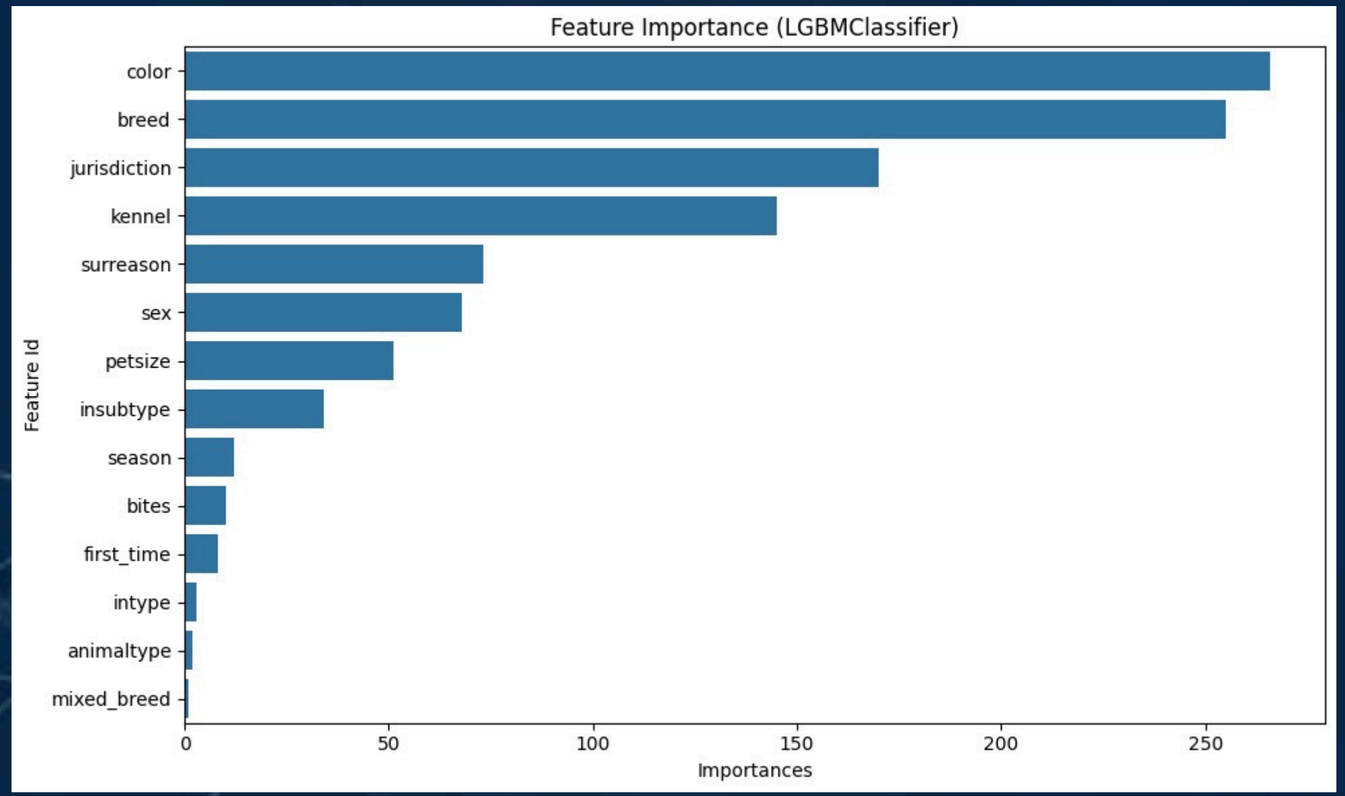
Confusion Matrix

Directly mapping the operational cost of False Positives vs. False Negatives.

Cross-Validation

Ensuring model stability and generalizability across diverse data folds.

Predictive Triage Methodology



Model Selection: LightGBM

- **Architectures Evaluated:** LightGBM, CatBoost, Random Forest, MLP, and Logistic Regression.
- **Optimization Engine:** Utilizing RandomizedSearchCV for rigorous hyperparameter tuning.
- **Why LightGBM?:** Highly efficient with categorical features and handles inherently imbalanced shelter data effectively via built-in weighted loss functions.

Model Performance Dashboard

	Model	PR AUC	Recall	Precision	F1-Score
0	LightGBM	0.522190	0.805556	0.286939	0.423152
1	CatBoost	0.513241	0.850000	0.253311	0.390306
2	Random Forest	0.446228	0.744444	0.275342	0.402000
3	MLP	0.429660	0.479630	0.405956	0.439728
4	Logistic Regression	0.348928	0.616667	0.260156	0.365934

K-Modes vs. Hierarchical Clustering

Comparative Analysis (Sample of 30,000 records)

- **Internal Cohesion (WSS):** K-Modes achieved a significantly lower/better score (164,638) compared to Hierarchical (172,742), proving it is more effective at grouping highly similar animals.
- **Cluster Separation (BSS)**: Hierarchical clustering created more distinct boundaries (0.654 vs K-Modes' 0.595), but these clusters had higher internal variance, making them less actionable.

Conclusion & Model Selection

K-Modes was selected as the final model because it achieved a higher overall Silhouette Score (0.159 vs 0.153). K-Modes also provides superior internal cohesion, better scalability for the full dataset, and aligns natively with the categorical shelter data.

High-Risk Profile & Mortality Rates

Highest Risk Profile Characteristics

- Cat
- Stray (OTC intake)
- Medium size
- Black color
- Neutered ('N')
- No prior bite history ('N')
- Non-mixed breed
- Intake during the Summer season

Mortality Rate per Cluster

The K-Modes algorithm successfully isolated three distinct risk groups, with mortality rates strictly ranked from highest to lowest risk.

Conclusion: A Holistic Digital Safety Net

- Combining supervised (LightGBM) for individual triage and unsupervised (K-Modes) for population strategy creates a robust, multi-layered safety net.
- This integrated framework empowers proactive resource allocation, enhances operational efficiency, and ultimately improves outcomes for all animals in the shelter.